

Units, System International

- 1 Length - m
- 2 Mass - kg
- 3 Time -s
- 4 Temperature - K
- 5 Force - N, $1N = 1kgm/s^2$
- 6 Energy - J
- 7 Power - rate of energy transfer or storage - W
=J/s, $1kW = 1000W$

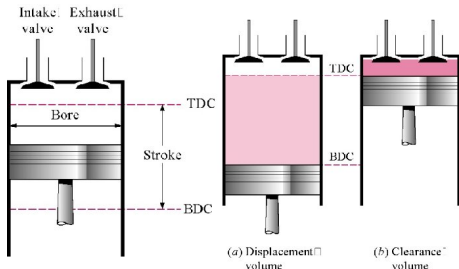
- 1 Pressure - Pa,
 $1Pa = 1N/m^2$
- 2 $1\text{ bar} = 10^5\text{ Pa} = 100\text{ kPa}$,
= 0.1 Mpa
- 3 $1\text{ atm} = 101.325\text{ kPa}$
- 4 Gauge pressure, Absolute pressure.
- 5 Density kg/m^3
- 6 Specific volume m^3/kg

Engine Dimensions

Dimensions

- 1 Bore = D
- 2 Stroke = TDC-BDC = L
- 3 Stroke Volume = $\pi L D^2 / 4$
- 4 Clearance Volume = $\pi X D^2 / 4$

Engine Cylinder



Engine piston and crank

Piston



Engine Crank



Air Standard Cycles

Thermodynamic Cycles

- 1 Process
- 2 Cycle
- 3 Representation of the cycle
- 4 First law applied to a process

Assumptions

- 1 Air is the working medium
- 2 Assumes ideal gas behaviour
- 3 Heat addition and removal externally
- 4 Processes are reversible

Nicolas Leonard Sadi Carnot

Father of Thermodynamics

- 1 June 1796 - 24 August 1832
- 2 Was a French military engineer and physicist
- 3 Published - Reflections on the Motive Power of Fire (1824)
- 4 Theorised the maximum efficiency of heat engines.

Nicolas Leonard Sadi Carnot

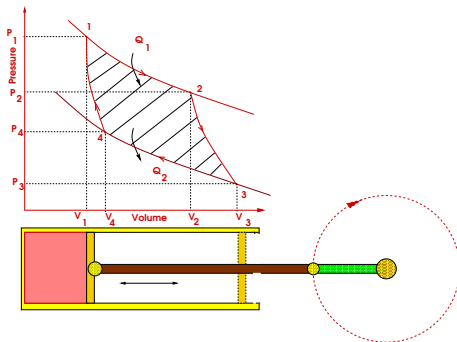


Carnot cycle

Assumptions

- 1 Proposed by Carnot in 1824
- 2 1-2 Rev. Isothermal heat addition
- 3 2-3 Rev. Adiabatic Expansion
- 4 3-4 Rev. Isothermal heat rejection
- 5 4-1 Rev. Adiabatic Compression

Carnot Cycle



Carnot cycle

Analysis

- 1 Heat Supplied rate $= Q_1$,
kJ/s
- 2 Heat Rejected rate $= Q_2$,
kJ/s
- 3 Work $= W = Q_1 - Q_2$
- 4 Efficiency
 $= \eta = [Q_1 - Q_2] / Q_1$.

Carnot Cycle

- 1 maximum efficiency
- 2 Standard for other cycles
- 3 Practical difficulties

Nikolaus Otto

Spark Ignition Engines

- 1 10 June 1832- 26 January 1891, Cologne
- 2 German engineer of the first internal-combustion engine
- 3 The four-stroke engine was first patented by Alphonse Beau de Rochas in 1861

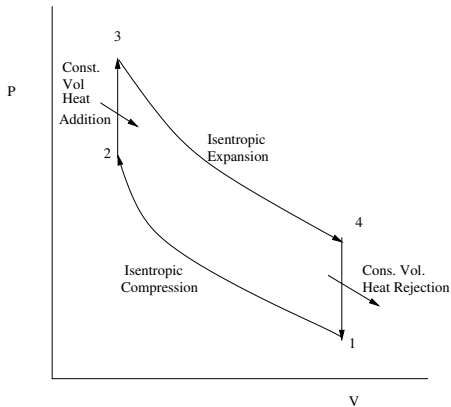
Nikolaus August Otto



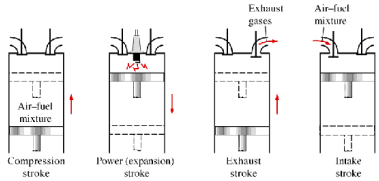
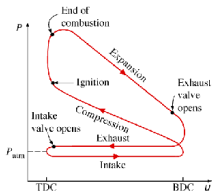
Otto Cycle

Processes

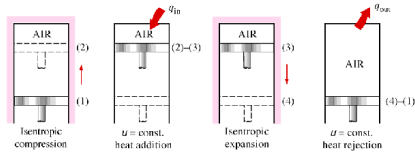
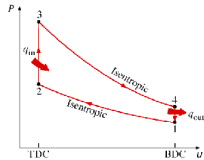
- 1 1-2 Rev. Adiabatic Compression
- 2 2-3 Const. Volume Heat Addition
- 3 3-4 Re. Adiabatic Expansion
- 4 4-1 Const. Volume Heat Rejection



Working of Otto Cycle



(a) Actual four-stroke spark ignition engine



(b) Ideal Otto cycle

a) Actual Otto Cycle working b) Ideal Otto Cycle working

Analysis

$$\text{Heat Supplied} = Q_1 = mc_v(T_3 - T_2)$$

$$\text{Heat Rejected} = Q_2 = mc_v(T_4 - T_1)$$

$$\text{Efficiency} = \eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{(T_4 - T_1)}{(T_3 - T_2)}$$

$$\frac{T_2}{T_1} = \left(\frac{v_1}{v_2}\right)^{\gamma-1}$$

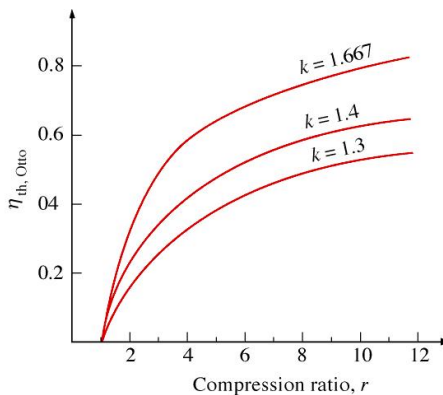
$$\frac{T_3}{T_4} = \left(\frac{v_4}{v_3}\right)^{\gamma-1} = \left(\frac{v_1}{v_2}\right)^{\gamma-1}$$

$$\frac{T_3}{T_2} - 1 = \frac{T_4}{T_1} - 1$$

$$\frac{(T_4 - T_1)}{(T_3 - T_2)} = \frac{T_1}{T_2} = \left(\frac{v_2}{v_1}\right)^{\gamma-1}$$

$$\eta_{\text{otto}} = 1 - \left(\frac{v_2}{v_1}\right)^{\gamma-1} = 1 - \frac{1}{r_k^{\gamma-1}}, r_k = \text{Compression ratio} = \frac{v_1}{v_2}$$

Effect of Compression Ratio



- 1 Higher Compression ratio $\Rightarrow$ Higher Efficiency.
- 2 Limited by Engine knock
- 3 γ depends on working fluid

Work Output

$$\text{Work output} = \frac{p_3 V_3 - p_4 V_4}{\gamma - 1} - \frac{p_2 V_2 - p_1 V_1}{\gamma - 1}$$

$$\frac{v_1}{v_2} = \frac{V_1}{V_2} = r_k, V_1 = V_2 r_k = V_4$$

$$\frac{p_2}{p_1} = \frac{p_3}{p_4} = \left[\frac{V_1}{V_2} \right]^\gamma = r_k^\gamma$$

$$\text{let } \frac{p_3}{p_2} = \frac{p_4}{p_1} = r_p$$

$$W_{net} = \frac{p_1 V_1}{\gamma - 1} \left[\frac{p_3 V_3}{p_1 V_1} - \frac{p_4 V_4}{p_1 V_1} - \frac{p_2 V_2}{p_1 V_1} + 1 \right]$$

$$= \frac{p_1 V_1}{\gamma - 1} \left[\frac{r_p r_k^\gamma}{r_k} - r_p - \frac{r_k^\gamma}{r_k} + 1 \right]$$

$$= \frac{p_1 V_1}{\gamma - 1} \left[r_p r_k^{\gamma-1} - r_p - r_k^{\gamma-1} + 1 \right]$$

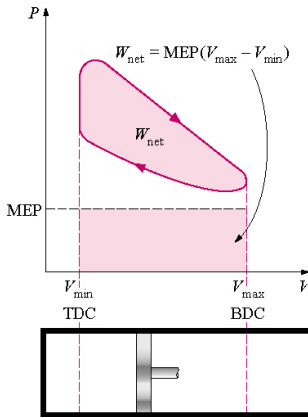
$$W_{net} = \frac{p_1 V_1}{\gamma - 1} (r_p - 1)(r_k^{\gamma-1} - 1)$$

Mean Effective Pressure

- 1 Average pressure with same work out put.
- 2 Work output / Swept volume
- 3 swept volume = $V_1 - V_2 = V_2(r_k - 1)$

$$4 \quad p_m = \frac{\frac{p_1 V_1}{\gamma - 1} (r_p - 1) (r_k^{\gamma - 1} - 1)}{V_2 (r_k - 1)}$$

$$5 \quad p_m = \frac{p_1 r_k (r_p - 1) (r_k^{\gamma - 1} - 1)}{(\gamma - 1) (r_k - 1)}$$

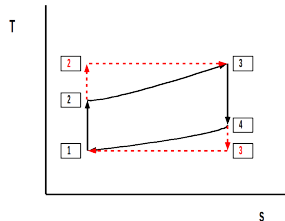


Comparison

- 1 Heat addition Const Temp Vs Const Vol
- 2 Heat rejection Const Temp Vs Const Vol
- 3 Otto efficiency will be less than carnot

Carnot - Otto

Carnot Otto comparison

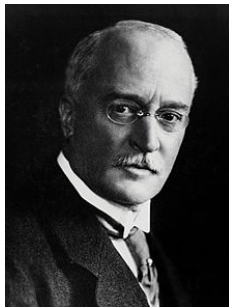


Rudolf Diesel

Compression Ignition Engines

- 1 Diesel was born at 38 Rue Notre Dame in Paris, France in 1858
- 2 Diesel received patent in 1893
- 3 Passed away in September 1913
- 4 Diesel engine required a heavier, more robust construction than a gasoline engine

Rudolf Diesel



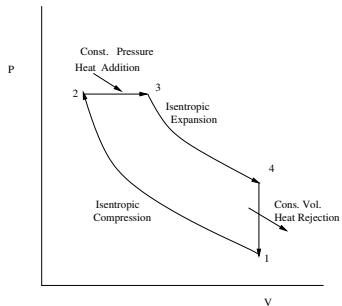
Diesel Cycle

Processes

- 1 1-2 Rev. Adiabatic Compression
- 2 2-3 Const. Pressure Heat Addition
- 3 3-4 Rev. Adiabatic Expansion
- 4 4-1 Const. Volume Heat Rejection

Diesel cycle

P-V Diagram



Analysis

$$\text{Heat Supplied} = Q_1 = mc_p(T_3 - T_2)$$

$$\text{Heat Rejected} = Q_2 = mc_v(T_4 - T_1)$$

$$\text{Efficiency} = \eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{c_v(T_4 - T_1)}{c_p(T_3 - T_2)}$$

$$\eta_{diesel} = 1 - \frac{(T_4 - T_1)}{\gamma(T_3 - T_2)}$$

$$\text{Compression, } r_k = \frac{V_1}{V_2} = \frac{v_1}{v_2}$$

$$\text{Expansion ratio, } r_e = \frac{V_4}{V_3} = \frac{v_4}{v_3}$$

$$\text{Cut off ratio, } r_c = \frac{V_3}{V_2} = \frac{v_3}{v_2}$$

$$r_k = r_e r_c$$

$$\frac{T_4}{T_3} = \left[\frac{v_3}{v_4} \right]^{\gamma-1} = \frac{1}{r_e^{\gamma-1}}$$

$$T_4 = T_3 \frac{r_c^{\gamma-1}}{r_k^{\gamma-1}}$$

Analysis

Process 2-3, $\frac{T_2}{T_3} = \frac{p_2 v_2}{p_3 v_3} = \frac{v_2}{v_3} = \frac{1}{r_c}$

$$T_2 = T_3 \frac{1}{r_c}$$

Process 1-2, $\frac{T_1}{T_2} = \left[\frac{v_2}{v_1} \right]^{\gamma-1} = \frac{1}{r_k^{\gamma-1}}$

$$T_1 = T_2 \frac{1}{r_k^{\gamma-1}} = \frac{T_3}{r_c r_k^{\gamma-1}}$$

Substituting values of T_1 , T_2 , T_4

$$\eta = 1 - \frac{T_3 \frac{r_c^{\gamma-1}}{r_k^{\gamma-1}} - \frac{T_3}{r_c r_k^{\gamma-1}}}{\gamma \left[T_3 - T_3 \frac{1}{r_c} \right]}$$

$$\eta_{Diesel} = 1 - \frac{r_c^{\gamma} - 1}{\gamma (r_c - 1) r_k^{\gamma-1}}$$

Diesel Compression Ratio

Compression Ratio

- 1 Efficiency increase with compression ratio.
- 2 Efficiency decreases with increase in cut off ratio.
- 3 Cut off ratio depends on the fuel spraying time.

Effect of Compression Ratio

Efficiency Vs Compression Ratio

